

Yarn Dyeing Quality Control

Inspection specification and app design
for three stage gates — raw material, post-dyeing, post-winding

Plant	2002 · Dyeing
System	KSD · S/4HANA · client 500
Throughput	~75 dye batches per day
Backend	Standard SAP QM inspection lots
Capture	Tablet-first shop-floor UI5 apps
Standards basis	ISO 105 series, ISO 2062, ISO 3071, ISO 2859-1, AATCC
Status	Draft for review — not yet approved
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Read section 2 first. This specification rests on four assumptions about your fibre, process and laboratory capability. If any of them is wrong, the characteristic tables and tolerance limits change materially. Nothing should be built until those four are confirmed.

1. What this document is

Three inspection stage gates for batch-wise package dyeing of filament yarn, specified to international test standards, and the three tablet applications that capture them. It covers what is measured, to which standard, against what limit, on how many samples, how often, and who decides.

It deliberately stops short of code. The characteristic tables in sections 5–7 are the part that needs your judgement as much as mine, and they are cheaper to argue about now than after they are built into master data.

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2. Assumptions that must be confirmed

I have inferred the process from your own data model rather than assuming a generic textile plant. Each inference below is stated with its evidence so you can correct it quickly.

Assumption	Evidence in your system	If wrong
A1 — Synthetic filament yarn , most likely polyester DTY/FDY, not staple or natural fibre	<code>ZZMARA</code> carries <code>ZZDENIR</code> (denier) and <code>ZZFILAM</code> (filament count). Denier and filament count are filament-yarn attributes; staple yarn is specified in <code>Ne/Nm</code> with a count and twist.	Cotton or blends would replace disperse-dye tests with reactive-dye tests, add mercerisation and singeing checks, and change every fastness limit. Roughly half of sections 5–6 would be rewritten.
A2 — Package (cheese) dyeing in HTHP vessels, not hank or beam	<code>ZPP_BATCHN</code> has a <code>CHEESES</code> quantity per batch. A cheese is a dye package.	Hank dyeing removes inside–outside package variation as a characteristic and changes the sampling unit from package to hank bundle.
A3 — The plant batch is the quality unit	<code>ZPP_BATCHN</code> links <code>GREY_CODE</code> → <code>DYE_CODE</code> with <code>AUFNR</code> , <code>BCHDATE</code> , <code>QTY</code> and <code>CHEESES</code> . Operators already work to this number.	If quality is judged per production order or per customer lot instead, the inspection lot anchor changes.
A4 — Winding is a separate downstream step producing the saleable cone	Separate packing tables (<code>ZPP_PACK</code>), a Packing Register, and winding referenced in the plant runbook (<code>ZJOBREPTN</code> — dyeing / winding job cards).	If winding is inline with dyeing, stages 2 and 3 merge into one gate.

A fifth unknown, and the most important one: what does your laboratory actually have? A specification that calls for xenon-arc light fastness or Classimat fault classification is worthless if the instruments are not on site. Section 14 lists the equipment each characteristic requires. Please mark that list before we go further — I would rather cut a characteristic than specify a test that quietly never gets done.

3. Process model and where the gates sit

Three gates, each with a different question to answer and a different cost of being wrong.

Gate	Material state	Question it answers	Cost of a miss	SAP anchor
Stage 1 Raw material	Greige / RFD yarn, undyed packages	Will this yarn dye level, and will it survive the dyebath?	Highest. A dye-affinity fault is invisible until after dyeing, and by then the batch is scrap or a re-dye.	ZPP_BATCHN.GREY_CODE , inspection type 01 at goods receipt
Stage 2 After dyeing	Dyed, dried packages	Is the shade right, is it level, is it fast, and is the yarn undamaged?	High. Detected here it is a re-dye; missed here it becomes a customer claim.	ZPP_BATCHN.DYE_CODE + AUFNR , inspection type 03 in-process
Stage 3 After winding	Wound cones, ready to pack	Is the package sound, and did winding damage the yarn or the shade?	Moderate but frequent. Winding faults are the commonest cause of running complaints at the customer's machine.	ZPP_PACK / HU, inspection type 04 at GR from production

Why stage 1 carries the most weight

This is the point most plants under-invest in, and it is where I would put the effort first. Dye uniformity in polyester is governed by properties fixed long before the yarn reaches you — draw ratio, heater profile, spin finish level, crystallinity. A package that differs in dye affinity from its neighbours produces barré that no amount of process control in the dyehouse can correct. The greige tube-dye test in section 5 costs perhaps twenty minutes and prevents a full batch re-dye, which at your volume is the single highest-return test in this document.

On the existing apps. The re-dye path already has SAP support in this system: *Cancel Production Confirmation* reverses the dyeing confirmation and *WIP Batch Close* reopens the batch. Section 9 wires the quality rejection decision into that existing loop rather than inventing a parallel one.

4. The throughput problem, and tiered testing

This is the constraint that shapes everything else, so it belongs before the characteristic tables rather than after them.

Seventy-five batches a day. The full post-dyeing battery in section 6 is twenty-two characteristics. Run in full on every batch that is **1,650 tests per day**, several of which take longer than a day to produce a result — ISO 105-B02 light fastness alone runs for tens of hours of xenon exposure. Any specification that implies testing everything on everything will be quietly abandoned within a fortnight, and you will be left with no quality record at all.

So the battery is tiered. Three tiers, applied per characteristic:

Tier	Frequency	Rationale	Daily volume at 75 batches
TIER 1 In-line	Every batch, 100%	Fast, cheap, catches the failures that actually occur day to day. Result available within the batch cycle.	~7 characteristics × 75 = 525 results
TIER 2 Shade qualification	First batch of each shade / recipe, and on any recipe change	Fastness is a property of the dye recipe, not of the individual batch. Testing it once per recipe is scientifically correct and commercially sufficient.	~8–12 new shades/day × 10 = ~100 results
TIER 3 Periodic & compliance	Weekly, monthly, per buyer lot, or per certification cycle	Regulatory and long-duration tests. Formaldehyde, azo amines, light fastness. Often sent to an external accredited laboratory.	Handful per week

The resulting load is manageable. Roughly 625 result entries per day. On a tablet designed for the job — batch chosen by scan, one tap per grey-scale rating, numeric pad for the rest — that is about 4–5 seconds per result, or **under an hour of total data entry across the day**. That is the design target for the apps in section 11, and it is the number to hold me to.

Tier 2 depends on a reliable shade/recipe identifier. The apps must know whether the batch in front of the technician is the first of its recipe today. `ZPP_BATCHN` carries `DYE_CODE` but I have not yet confirmed whether the dye *recipe* is separately identified — the Recipe Master app suggests it is. This needs settling before tier 2 can be automated; until then it falls to the technician to flag, which is less reliable.

5. Stage 1 — Raw material inspection before dyeing

Applied to incoming greige / RFD filament yarn, per supplier lot and delivery. Sampling per section 8.

5.1 Yarn physical characteristics

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
1	Linear density	ISO 2060 ASTM D1907	Quant	dtex	nominal $\pm 2.0\%$	1	Skein method, 100 m wraps, conditioned
2	Tenacity	ISO 2062 ASTM D2256	Quant	cN/dtex	≥ 3.8 (DTY) ≥ 4.0 (FDY)	2	CRE, 500 mm gauge, 20 breaks
3	Elongation at break	ISO 2062	Quant	%	18–28	2	Same test as #2
4	Elongation CV%	ISO 2062	Quant	%	≤ 8.0	2	Variability matters more than the mean for levelness
5	Boiling water shrinkage	ASTM D2259	Quant	%	5.0–9.0	1	100°C, 30 min. Key predictor of barré
6	Crimp contraction (DTY)	DIN 53840-1	Quant	%	18–30	2	Omit for FDY
7	Evenness U%	ISO 16549	Quant	%	≤ 1.2	2	Uster tester
8	Interlace / nips	ASTM D4724	Quant	nips/m	per grade spec	2	Affects package build and unwinding
9	Broken filaments	Visual under tension	Quant	count/kg	≤ 2	1	10-minute run over black board

5.2 Dye-readiness characteristics HIGHEST VALUE

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
10	Dye uniformity (tube / sock test)	ISO 105-A02 grey scale	Qual	grade	≥ 4	1	Knit a tube from n packages, dye together with a critical disperse dye, rate variation. The single most valuable test at this gate.
11	Oil pick-up / spin finish	Solvent extraction	Quant	%	0.8–1.2	2	High or uneven OPU causes uneven scouring and spotting
12	Moisture regain	ISO 6741-1	Quant	%	≤ 0.4	2	Oven dry 105°C to constant mass
13	Whiteness index (for optical white / pastel base)	ISO 105-J02 CIE WI	Quant	WI	≥ 70	2	Only where pastel or white shades are planned
14	pH of aqueous extract	ISO 3071	Quant	pH	5.5–7.5	3	Carried-over alkali disturbs disperse dyeing

5.3 Package and identity characteristics

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
15	Package weight	—	Quant	kg	nominal $\pm 3\%$	1	Drives dyebath liquor ratio
16	Package hardness / density	Durometer	Quant	° Shore	28–34	1	Critical. Hard packages resist liquor flow → inside–outside variation
17	Package build defects	Visual	Qual	P/F	Pass	1	Ribboning, bulge, soft or hard edge, damaged tube
18	Contamination / foreign fibre	Visual	Qual	P/F	Pass	1	Oil, rust, coloured fibre, dust

19	Lot identity vs SAP batch	—	Qual	P/F	Pass	1	Supplier tag vs GREY_CODE / batch. Merged lots are a common root cause of barré
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Merged lots deserve their own rule. Characteristic 19 is not clerical. Dyeing a vessel loaded from two supplier lots — or two draw-texturing positions — is one of the most reliable ways to produce barré in polyester. Your *Merge Details* app suggests merging is a recognised operation here. I would make single-lot loading a hard rule and require an explicit override, recorded against the batch, when it is broken.

6. Stage 2 — Inspection after dyeing

Applied per dye batch after dyeing and drying, before release to winding. This is the gate the customer ultimately judges you on.

6.1 Shade and levelness EVERY BATCH

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
1	Shade vs approved standard ΔE CMC(2:1)	ISO 105-J03 AATCC EP6	Quant	ΔE	≤ 1.0	1	Spectrophotometer, D65/10°, specular included, UV calibrated. Wind a flat card to constant thickness
2	Metamerism index	ISO 105-J03	Quant	MI	≤ 0.5	2	D65 / TL84 / A. Recipe property, not batch property
3	Inside–outside package variation	ISO 105-A02	Qual	grade	$\geq 4-5$	1	Unwind inner and outer layers of the same cheese. The characteristic package dyeing fails on
4	Package-to-package within batch	ISO 105-A02	Qual	grade	≥ 4	1	Top, middle, bottom of the carrier
5	Batch-to-batch vs master	ISO 105-A02	Qual	grade	≥ 4	1	Against the retained master for that shade
6	Levelness / barré visual	ISO 105-A02	Qual	grade	≥ 4	1	Knitted sock under D65 in a light cabinet
7	Stains, tip-dyeing, rust marks	Visual	Qual	P/F	Pass	1	Tip-dyeing indicates carrier or flow problems

6.2 Colour fastness PER RECIPE, NOT PER BATCH

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
8	Washing — colour change	ISO 105-C06 A1S	Qual	grade	≥ 4	2	Grey scale ISO 105-A02
9	Washing — staining	ISO 105-C06 A1S	Qual	grade	$\geq 3-4$	2	Multifibre adjacent, grey scale ISO 105-A03
10	Rubbing dry	ISO 105-X12	Qual	grade	≥ 4	2	Crockmeter
11	Rubbing wet	ISO 105-X12	Qual	grade	≥ 3	2	Most frequent fastness failure in deep shades
12	Perspiration acid	ISO 105-E04	Qual	grade	≥ 4	2	
13	Perspiration alkaline	ISO 105-E04	Qual	grade	≥ 4	2	
14	Water	ISO 105-E01	Qual	grade	≥ 4	2	
15	Sublimation / dry heat	ISO 105-P01	Qual	grade	≥ 4	2	180°C / 30 s. Specific to disperse dyes on polyester — do not omit
16	Hot pressing	ISO 105-X11	Qual	grade	≥ 4	2	
17	Light	ISO 105-B02	Qual	blue wool	≥ 4	3	Xenon arc. Runs for days — schedule, never gate on it

6.3 Yarn condition and chemical safety

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
18	Tenacity retention vs greige	ISO 2062	Quant	%	≥ 92	2	Detects thermal or hydrolytic damage in the dyebath
19	pH of aqueous extract	ISO 3071	Quant	pH	4.5–7.5	1	Cheap, fast, and a direct indicator of wash-off adequacy
20	Residual moisture after drying	ISO 6741-1	Quant	%	≤ 0.5	1	

							Wet packages mildew and wind badly
21	Residual spin finish after scouring	Solvent extraction	Quant	%	≤ 0.15	3	Incomplete scouring causes patchy shade
22	Free formaldehyde	ISO 14184-1	Quant	ppm	≤ 75	3	OEKO-TEX / buyer requirement
23	Banned azo amines	EN ISO 14362-1	Quant	ppm	< 20	3	REACH. Normally external accredited lab, per certification cycle

Why fastness is tier 2 and not tier 1. Colour fastness is a property of the dye chemistry and the wash-off process, not of the individual batch. Two batches of the same recipe run correctly will not differ in fastness. Testing it once per recipe — and re-testing whenever the recipe, dye lot or supplier changes — is both scientifically sound and defensible in a buyer audit, provided the linkage from batch to qualified recipe is recorded. That linkage is what the app must guarantee.

7. Stage 3 — Inspection after winding

Applied to wound cones before packing. Winding does not change the dye, but it readily damages the yarn and the package, and it can abrade the surface enough to shift apparent shade.

7.1 Package geometry and build

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
1	Net package weight	—	Quant	kg	nominal $\pm 1.5\%$	1	Commercial exposure — short weight is a claim
2	Package length / metrage	ISO 2060	Quant	m	nominal $\pm 2\%$	2	Where sold by length
3	Package density	—	Quant	g/cm ³	0.42–0.50	1	Too soft collapses in transit, too hard will not unwind
4	Cone taper and traverse	—	Qual	P/F	Pass	1	Against the agreed cone format
5	Ribboning / patterning	Visual	Qual	P/F	Pass	1	Causes sloughing at the customer
6	Nose bulge, soft or hard edge	Visual / durometer	Qual	P/F	Pass	1	
7	Sloughing-off / stitching	Unwinding trial	Qual	P/F	Pass	2	Overend unwind at production speed

7.2 Yarn quality after winding

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
8	Splice strength retention	ISO 2062	Quant	%	≥ 80	2	Tested on the spliced length vs parent yarn
9	Splice appearance	Visual	Qual	grade	≥ 4	1	Bulk and visibility at the splice
10	Splices per package	Count	Quant	count	≤ 3	1	Tighten for critical customers
11	Yarn fault classification	ASTM D6197 Uster Classimat	Quant	faults/ 100 km	per class limits	3	Only if a Classimat is available
12	Broken filaments	Visual under tension	Quant	count/kg	≤ 2	1	Compare against the stage 1 figure — the delta is what winding did
13	Hairiness increase vs pre-wind	Uster H	Quant	ΔH	≤ 0.3	2	Direct measure of winding-induced abrasion
14	Winding tension CV%	Tensiometer	Quant	%	≤ 6	2	Per winding position — also a machine-maintenance signal

7.3 Final release checks

#	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Note
15	Shade re-check after winding	ISO 105-A02	Qual	grade	$\geq 4-5$	1	Abrasion can lighten the surface, especially in deep shades
16	Contamination / oil stain from winder	Visual	Qual	P/F	Pass	1	
17	Moisture at packing	ISO 6741-1	Quant	%	≤ 0.5	1	
18	Label / barcode vs SAP HU	—	Qual	P/F	Pass	1	Scan-verified against the handling unit

8. Sampling plans

Attribute characteristics follow **ISO 2859-1**, single sampling, General Inspection Level II, normal severity. Variable characteristics use a fixed sample per batch as noted.

8.1 Attribute sampling — ISO 2859-1, GIL II, normal

Lot size (packages)	Code letter	Sample size	Ac / Re at AQL 2.5 (major)	Ac / Re at AQL 1.0 (critical)
26 – 50	D	8	0 / 1	0 / 1
51 – 90	E	13	1 / 2	0 / 1
91 – 150	F	20	1 / 2	1 / 2
151 – 280	G	32	2 / 3	1 / 2
281 – 500	H	50	3 / 4	1 / 2

AQL assignment: critical (safety, chemical compliance, identity) 1.0 · major (shade, fastness, strength, package integrity) 2.5 · minor (cosmetic) 4.0. The lot size is the cheese or cone count from `ZPP_BATCHN.CHEESES` or the packing record, so the app can derive the sample size automatically rather than asking the technician to look it up.

8.2 Variable sampling — fixed per batch

Characteristic group	Sample	Positioning
Shade ΔE	3 packages per batch	Top, middle and bottom of the carrier — deliberately chosen to expose vertical liquor-flow gradient
Inside–outside variation	2 packages, 2 layers each	Innermost and outermost 50 g of each
Tensile (ISO 2062)	20 breaks	Across at least 5 packages
pH, moisture	1 composite	Blended from the shade sample packages
Fastness battery	1 composite	Representative of the qualifying batch for the recipe

Position matters more than count. In package dyeing the dominant failure mode is a gradient — inside to outside, and top to bottom of the carrier. Three samples chosen to span that gradient tell you far more than ten chosen at random. The apps should therefore prompt for *position*, not just for a package number, and record it against the result.

9. Decision rules and the re-dye loop

9.1 Usage decision codes

Code	Meaning	Trigger	Stock / process consequence
A	Accept	All tier-1 characteristics within limits; recipe holds a valid tier-2 qualification	Unrestricted use; batch released to next stage
A1	Accept with deviation	Marginal result, concession signed by QC Manager, customer notified where the contract requires it	Unrestricted; deviation note carried on the batch
RD	Re-dye / correct	$\Delta E > 1.0$ but shade correctable; unlevelness $\geq$ grade 3	Blocked; batch reopened for correction — see 9.2
ST	Strip and re-dye	Shade not correctable by topping; severe barré	Blocked; stripping process; strength re-check mandatory afterwards
DG	Downgrade	Cosmetic or minor faults; yarn sound	Restricted use; B-grade stock
R	Reject	Tenacity retention $< 92\%$; chemical compliance failure; identity failure	Blocked; scrap or supplier return decision

9.2 The re-dye loop, using apps you already have

An **RD** or **ST** decision at stage 2 must reopen the batch for correction. That path exists in this system already and should not be rebuilt:

- 1 Post-Dyeing QC app records the usage decision **RD** and blocks the batch.
- 2 The batch must be reopened before the confirmation can be reversed — this is the dependency the plant runbook records.
WIP Batch Close app, Reopen action, reason code **QC-REDYE**.
- 3 **Cancel Production Confirmation** reverses the dyeing confirmation, writing the reversal to **AFRU**.
- 4 Correction is dyed; a new inspection lot is raised automatically on re-confirmation.
- 5 Re-dyed batches carry a **mandatory tenacity re-check** — each thermal cycle costs strength, and two corrections on the same batch is normally the point to stop.

Rule worth adopting: two corrections maximum. After a second re-dye, require a QC Manager signature to attempt a third. Polyester tolerates repeated HT cycles better than most fibres, but tenacity and hand deteriorate, and the economics have usually gone negative well before the quality does.

10. SAP QM design

10.1 Inspection types and origins

Stage	Insp. type	Origin	Lot created by	Anchor
Stage 1	01	Goods receipt against purchase order	Automatic at GR	Material = GREY_CODE , supplier batch
Stage 2	03	In-process, production order	Automatic at order release	AUFNR + BATCHNO
Stage 3	04	Goods receipt from production	Automatic at GR of wound stock	Material = packed cone, HU

10.2 Characteristic types

MIC type	Count	Capture control	Valuation
Quantitative with limits	~26	Numeric keypad, spec limits shown inline	Automatic against tolerance
Qualitative — grey scale	~17	Nine-position selector: 1, 1-2, 2, 2-3, 3, 3-4, 4, 4-5, 5	Selected set, threshold per characteristic
Qualitative — pass/fail	~14	Two-state toggle, defect code on fail	Selected set
Qualitative — blue wool	1	Eight-position selector, 1–8	Selected set

10.3 Backend objects to be built

Object	Type	Purpose
ZI_QC_INSP_LOT	CDS custom entity	Worklist over QALS joined to ZPP_BATCHN , so the technician sees the plant batch number and shade, not just a lot number
ZI_QC_INSP_CHAR	CDS custom entity	Characteristics for a lot from QAMV , with recorded results from QAMR and the spec limits
ZCL_QC_INSP_LOT_QUERY	Query class	if_rap_query_provider~select . Same pattern as the four existing query classes, including get_search_expression() and CompanyCode filtering
ZBP_I_QC_INSPECTION	Behaviour pool	Unmanaged. Actions recordResults , saveDraft , setUsageDecision
ZQC_RESULT_DRAFT	Z table	Draft buffer — see the note below

BAPIs used

List open lots	BAPI_INSPL0T_GETLIST
Read characteristics	BAPI_INSP0PER_GETDETAIL / BAPI_INSPCHAR_GETREQUIREMENTS
Write one result	BAPI_INSPCHAR_SETRESULT
Close the operation	BAPI_INSP0PER_RECORDRESULTS
Usage decision	BAPI_INSPL0T_SETUSAGEDECISION
Commit	BAPI_TRANSACTION_COMMIT – via DESTINATION 'NONE' , per the pattern established in the confirmation-cancel app

Why a draft table. A fastness battery is not entered in one sitting — a technician starts a lot, the perspirometer runs for four hours, they come back. QM has no draft concept for results: a result is either recorded or it is not. ZQC_RESULT_DRAFT holds partial entry keyed by lot, operation, characteristic and sample, and is cleared when BAPI_INSP0PER_RECORDRESULTS succeeds. Without it the technician must either hold results on paper or record them prematurely, and both defeat the purpose.

10.4 Fiori objects

App	BSP	Component ID	Semantic object / action	Service binding
Raw Material QC	ZQC_RAW_DYE	kejriwal.qm.qcrawdyeing	QcRawMaterial / inspectKejriwal	ZUI_QC_RAW_DYEING_04

Post-Dyeing QC	ZQC_POST_DYE	kejriwal.qm.qcpostdyeing	QcPostDyeing / inspectKejriwal	ZUI_QC_POST_DYEING_04
Post-Winding QC	ZQC_POST_WIND	kejriwal.qm.qcpostwinding	QcPostWinding / inspectKejriwal	ZUI_QC_POST_WINDING_04

Package ZKGPL_FIORI , not \$TMP . Twenty of the twenty-five apps already built in this landscape sit in \$TMP and cannot be transported to QSA. These three must not join them. The package is set in ui5-deploy.yaml before the first deployment, and it is far easier to get right at the start than to correct afterwards.

11. Application design

Three apps, one pattern. A lab technician standing at a bench, possibly gloved, working through batches at pace. Every design decision below follows from that.

11.1 Screen 1 — Worklist

Post-Dyeing QC — Open Inspections (18)

FILTER BAR — pinnable

Plant 2002 Date Today Batch [scan] Shade Machine Status Open [Go]

B-2026-04417 Navy 19-4024 HT-3 220 kg 48 cheeses
Due 14:30 · 7 of 7 characteristics open · **FIRST OF RECIPE — full battery**

B-2026-04418 Black 19-4005 HT-1 260 kg 56 cheeses
Due 15:10 · 3 of 7 recorded · draft saved 13:52

B-2026-04419 Beige 13-0917 HT-5 180 kg 40 cheeses
Due 15:45 · not started

Design decisions. The plant batch number leads, because that is what is written on the carrier tag — the QM inspection lot number is secondary and shown only in the detail. Shade name and machine are on the row because technicians triage by them. The *first of recipe* flag is calculated server-side and drives which characteristics appear, so tier 2 is enforced by the system rather than remembered by a person.

11.2 Screen 2 — Result capture

B-2026-04417 — Navy 19-4024 — HT-3 — 48 cheeses

1 Shade ΔE CMC(2:1) vs standard ISO 105-J03 · spec ≤ 1.00
Position [Top] [Middle] [Bottom] **0.74** ✓ **PASS**

3 Inside–outside variation ISO 105-A02 · min 4-5
1 1-2 2 2-3 3 3-4 4 **4-5** 5 ✓

6 Levelness / barré ISO 105-A02 · min 4
1 1-2 2 2-3 **3** 3-4 4 4-5 5 **✗ FAIL** defect code required

19 pH of aqueous extract ISO 3071 · spec 4.5–7.5
6.8 ✓ **PASS**

Save Draft Record Results Usage Decision →

Design decisions. The standard and the spec limit sit on every row, so the technician never has to remember a limit or consult a folder. Valuation is immediate and colour-coded — a fail is visible the moment it is entered, not after posting. Grey-scale ratings are a nine-position tap strip, never a dropdown; that is the difference between one second and five per result, and across 625 results a day it is the difference between the app being used and being abandoned. A failing characteristic demands a defect code before the lot can be posted, which is what makes the Pareto analysis in 11.4 possible.

11.3 Screen 3 — Usage decision

Summary of all characteristics with their valuations, the proposed decision derived from the rules in section 9, and the technician's confirmation or override. An override requires a reason. **RD** and **ST** decisions display the re-dye loop from 9.2 as an explicit next-step prompt, including the reason code to use in WIP Batch Close.

11.4 Deliberately out of scope for v1

Feature	Why not now
Spectrophotometer integration	Reading ΔE directly from the instrument removes transcription error and is the obvious v2. It needs the make and model, and a decision on file drop versus direct interface.
Defect Pareto dashboard	Worth building, but only once several weeks of coded defect data exist. Building it first produces an empty chart.
Photo capture on failure	Valuable for barré and stains. Needs a decision on attachment storage and archiving before it is added.
SPC control charts on ΔE	The right long-term use of this data — trend detection before a limit is breached. Requires a stable data history first.

12. Master data to be created

This is the substantial part of the work, and it is QM configuration rather than development. It cannot be done by a developer alone — it needs the QC manager and a QM consultant together.

Object	Count	Transaction	Notes
Master inspection characteristics	~58	QS21	19 stage 1, 23 stage 2, 18 stage 3. Some shared across stages — pH, moisture, tenacity — should be created once and reused
Code groups / catalogs	~6	QS41	Grey scale (9 codes), blue wool (8), pass/fail, plus defect catalogs per stage
Selected sets	~6	QS51	With valuation thresholds per characteristic
Inspection plans	3 × material group	QP01	One per stage, per material group
Sampling procedures	~4	QDV1	ISO 2859-1 attribute plans plus fixed-sample variable plans
Dynamic modification rules	2	QDR1	This is what implements tiering. Skip lots for tier-2 characteristics once the recipe is qualified
Usage decision selected set	1	QS51	Catalog 3, the six codes in section 9.1
Material master QM views	per material	MM02	Activate inspection types 01, 03, 04 with the right plans

Realistic effort. The 58 MICs are perhaps three days of careful data entry, but agreeing the tolerance limits will take longer than entering them, and that agreement is the valuable part. I would budget a working session per stage with whoever currently signs off quality, using the tables in sections 5–7 as the agenda.

13. Build plan

Phase	Work	Owner	Depends on	Output
0	Confirm assumptions A1–A4 and the laboratory equipment list	QC Manager	—	This document, corrected
1	Agree characteristics and tolerance limits, stage by stage	QC Manager + QM consultant	Phase 0	Approved characteristic tables
2	Create MICs, catalogs, selected sets, sampling procedures, plans	QM consultant	Phase 1	Master data in KSD
3	Activate inspection types on a pilot material; verify lots are created	QM consultant	Phase 2	Live inspection lots
4	Build Post-Dyeing QC end to end as the pattern app	Development	Phase 3	One working app
5	Pilot on one dyeing line for two weeks	QC Manager	Phase 4	Corrections to limits and UX
6	Replicate for Raw Material and Post-Winding	Development	Phase 5	Three working apps
7	Tiles, roles, transport to QSA	Basis	Phase 6	Production release

Phase 4 before phases 6. The lesson from the last three apps built in this landscape is that a pattern proven once is cheap to replicate, and a pattern replicated before it is proven is expensive to correct three times over. Two weeks of real use by a real technician will change the tolerance limits and the screen layout in ways neither of us can predict from here.

14. What I need from you

14.1 The four assumptions

Section 2, items A1–A4. A yes or a correction on each.

14.2 Laboratory capability

Please mark each: **in-house**, **external lab**, or **not available**. Where an instrument is not available the characteristic must be cut or outsourced — it should not sit in a specification as an aspiration.

Instrument	Enables	Characteristics affected
Spectrophotometer (D65/10°, UV)		Shade ΔE , metamerism, whiteness index — the core of stage 2
Light cabinet, multi-illuminant		All visual grey-scale assessments
Grey scales ISO 105-A02 / A03		17 qualitative characteristics
Tensile tester (CRE)		Tenacity, elongation, splice strength, retention
Crockmeter		Rubbing fastness dry / wet
Perspirometer + oven		Perspiration, water, sublimation, hot pressing
Launderometer / washing fastness		ISO 105-C06
Xenon-arc light fastness tester		ISO 105-B02 — commonly outsourced
Uster evenness tester		U%, hairiness
Uster Classimat		Yarn fault classification — cut if absent
Oven / moisture balance		Moisture regain, residual moisture
pH meter		ISO 3071 both stages
Durometer		Package hardness
Soxhlet / OPU extraction		Spin finish, residual finish

14.3 Commercial and compliance

- Which certifications are you held to — OEKO-TEX Standard 100, GRS, ZDHC, buyer-specific codes of conduct?
- Do any buyers impose their own fastness minima that override section 6? If so, those specifications supersede my proposed limits and I should encode theirs.
- Is the dye **recipe** separately identified in SAP, or only the dyed material code? Tier 2 automation depends on this.
- Who signs a usage decision, and who may approve an **A1** concession? This becomes an authorization design point.

14.4 Process detail

- Are batches ever loaded from more than one supplier lot? If so, how is that recorded today?
- What is the current re-dye rate? It sets the expected load on the loop in 9.2 and tells us where the tolerance limits should sit initially.
- Is winding always in-house, or sometimes subcontracted?

Appendix A — Standards register

Standard	Title / scope	Used at
ISO 105-A02	Grey scale for assessing change in colour	Dye uniformity, levelness, all shade-variation ratings
ISO 105-A03	Grey scale for assessing staining	Washing fastness staining
ISO 105-B02	Colour fastness to artificial light: xenon arc	Stage 2 tier 3
ISO 105-C06	Colour fastness to domestic and commercial laundering	Stage 2 tier 2
ISO 105-E01	Colour fastness to water	Stage 2 tier 2
ISO 105-E04	Colour fastness to perspiration	Stage 2 tier 2
ISO 105-J02	Instrumental assessment of relative whiteness	Stage 1, pastel and white bases
ISO 105-J03	Calculation of colour differences	ΔE CMC, metamerism index
ISO 105-P01	Colour fastness to dry heat (excluding pressing)	Sublimation — critical for disperse dyes
ISO 105-X11	Colour fastness to hot pressing	Stage 2 tier 2
ISO 105-X12	Colour fastness to rubbing	Stage 2 tier 2
ISO 2060	Linear density of yarn from packages (skein method)	Stage 1, stage 3 metrage
ISO 2062	Tensile properties of single-end yarns (CRE)	Stages 1, 2, 3
ISO 2859-1	Sampling procedures for inspection by attributes	All attribute sampling
ISO 3071	Determination of pH of aqueous extract	Stages 1 and 2
ISO 6741-1	Determination of mass after oven drying (moisture)	Stages 1, 2, 3
ISO 14184-1	Determination of formaldehyde, free and hydrolysed	Stage 2 tier 3
ISO 16549	Unevenness of textile strands — capacitance method	Stage 1
EN ISO 14362-1	Determination of certain aromatic amines from azo colorants	Stage 2 tier 3, REACH
ASTM D1907	Linear density of yarn (alternative to ISO 2060)	Stage 1
ASTM D2256	Tensile properties of yarns, single-strand method	Stage 1
ASTM D2259	Shrinkage of yarns	Boiling water shrinkage
ASTM D4724	Entanglement of filament yarns (interlace)	Stage 1
ASTM D6197	Short-term yarn evenness / fault classification	Stage 3
DIN 53840-1	Determination of crimp of textured filament yarns	Stage 1, DTY only
AATCC EP6	Instrumental colour measurement	Alternative reference for ΔE

Prepared as a design proposal for review. Tolerance limits are industry-typical values for polyester package dyeing and are **proposals, not requirements** — every one of them should be confirmed against your own process capability and your buyers' specifications before it is entered into SAP master data. A limit the plant cannot meet is worse than no limit at all, because it teaches operators to ignore the system.